

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

C04: *Examine intermediate code generation techniques to enhance code optimization (BL4)*

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

CLASS TEST

Code of Conduct:

I Athavan K(192524036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student :Athavan k

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization (15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

Q1. Construct three address code for the following

if((a < b) and ((c < d) or (a > d))) then

z=x + y *z

else

z = z + 1

Q2.Translate (a < b) and (c < d) or (e < f) into three address statements using back patching.

Q3. Consider the arithmetic expression:

$$A = (-A + B) * ((-A + B) * C)$$

(i).Analyze the given expression and explain how operator precedence and associativity influence its evaluation order.

(ii). Construct the Three Address Code (TAC) for the expression by breaking it into intermediate steps using temporary variables.

(iii). Represent the generated TAC in the following forms:

(a).Quadruples (op, arg1, arg2, result)

(b).Triples (index, op, arg1, arg2)

(c).Indirect Triples (pointer-based representation)

(iv). Compare the representations and analyze:

(a).Memory usage

(b).Ease of code optimization

(c).Advantages of indirect triples over triples

Q4. Construct three address code using the following production for Boolean expression $(a > b) \text{ or } (c < d) \text{ or } (e < f)$

$B \rightarrow B1 \parallel B2$

$B \rightarrow B1 \ \&\& \ B2$

$B \rightarrow ! \ B1$

$B \rightarrow E1 \ \text{relop} \ E2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

Q5. Generate the three address code for the following program fragment-

while $(A < C \text{ and } B > D)$ do

if $A = 1$ then $C = C + 1$

else while $A \leq D$ do

$A = A + B$

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

Construct three address code,

if $(a < b)$ and $(c < d)$ (or) $(a > d)$ then

$z = x + y * z$

else

$z = z + 1$

conditions : $a < b$
 $c < d$
 $a > d$

38/40

Labels : 21, 22, 23, 24, 25, 26

21:

if $a < b$ goto 22

goto 25

22:

if $c < d$ goto 24

goto 23

23:

if $(a > d)$ goto 24

goto 25

three address code

$24 \Rightarrow z = x + y * z$

$\therefore 24:$

$t_1 = y * z$

$t_2 = x + t_1$

$z = t_2$

$$L5 \Rightarrow 2 = 2 + 1$$

L5:

$$t_3 = 2 + 1$$

$$2 = t_3$$

2

Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three address statements using back patching

$[(a < b) \text{ AND } (c < d)] \text{ or } (e < f)$

if $(a < b)$ goto 3

goto 5

if $(c < d)$ goto 7

goto 5

if $(e < f)$ goto 7

goto 8

7. True

8. False

3

Construct three address code. Using the following production for boolean expression $(a < b)$ or $(c < d)$ or $(e < f)$

$$B \rightarrow B_1 \mid B_2$$

$$B \rightarrow B_1 \& B_2$$

$B \rightarrow 101$

$B \rightarrow E$, repeat 02

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

21: if ($a > b$) goto 2 true
goto 22

22: if ($c < d$) goto 2 true
goto 23

23: if ($e \neq f$) goto 2 true
goto 2 false

4

Generate the three address code for the following program fragment, while ($A \neq C$ and $B \neq D$) do

if $A = 1$ then $C = C + 1$
else while $A \neq D$ do
 $A = A + B$

21:

if $A \neq C$ goto 22
goto 2 END

L2::

if B > D goto L3
goto 1END

L3::

if A == 1 goto L4
goto L5

L4::

B = C + 1

C = + 1

goto L1

L5::

if A <= D goto L6
goto L1

L6::

X₂ = A + B

A = X₂

goto L5

2END